

IZELMADENE ACHRAF

Engineering Student - Embedded Systems & IoT

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PROFILE

Seeking a 6-month final-year internship starting February 2027 in software and hardware design of embedded systems. Passionate about innovation, I aim to apply my skills in C/C++ development, IoT integration, Real-Time Operating Systems (RTOS), and modeling to your R&D projects.

EXPERIENCE

Design Office Intern

Core Engineering Solutions - Berrechid, Casablanca Jul. - Aug. 2026

- Sizing of an MV/LV substation and integration of a self-consumption solar PV installation.
- Creation of single-line diagrams, power balances, and calculation notes (AutoCAD, Caneco BT, PVsyst).

Industrial Production Intern

Menara Prefa - Marrakech Jul. 2024

- Migration of an industrial PLC to Siemens S7 using TIA Portal, followed by preventive/corrective maintenance.
- Diagnosis of electrical faults and participation in the continuous improvement of industrial processes.

PROJECTS

Precision Agriculture IoT Ecosystem

- Deployment of a LoRa wireless IoT network connecting sensor nodes (Arduino) to a gateway (ESP32).
- Integration of an Edge AI (TFLite) and development of a web dashboard for global system supervision.

Predictive Maintenance 4.0 via AutoML

- Comparison of AutoML tools TPOT vs. Auto-WEKA on IoT data (vibration, temperature).
- Time-series cross-validation, SMOTE, automated ML pipelines, and NDT study via oil analysis.

Dimmer Control Prototype

- Design and simulation (Simulink/Proteus) of a power control circuit.
- Interfaced with Arduino (Hardware-in-the-Loop) to generate control signals.

EDUCATION

Engineering Student in Electrical Engineering and Smart Systems

Moroccan School of Engineering Sciences (EMSI) - Marrakech 2025 - Present

Electrical Networks and Installations, Electrical Safety, Power Electronics, Supervision & Digital Twin, Project Management...

Bachelor in Physical Sciences, Control Systems option

Semlalia Faculty of Sciences - Marrakech 2020 - 2024

Electromagnetism, Electronics, Signal Processing, Control Systems, Instrumentation, Industrial IT...

SKILLS

Engineering & Tools : Caneco BT, PVsyst, AutoCAD, MATLAB, Simulink, KiCad, 3DEXPERIENCE, Git, Docker
Programming & AI : Python, C/C++, JavaScript, HTML/CSS, Machine Learning, Scikit-learn, AutoML, MLOps
Soft Skills : Complex Problem Solving, Analytical Thinking, Autonomy, Adaptability, Teamwork

CERTIFICATIONS

- **Foundations of IoT Systems and Industrial Automation** — LearnQuest (Coursera)
- **Sensors and Sensor Circuit Design** — University of Colorado Boulder (Coursera)
- **Certified 3DEXPERIENCE Collaborative Industry Innovator** — Dassault Systèmes
- **MATLAB / Simulink Onramp** — MathWorks

ACTIVITIES & COMPETITIONS

- **1st Prize** -- Hack The Intelligence (AISEC Summit 2026)
- **2nd Prize** -- Vision AI Competition (EMSI 2026)
- **Scientific Presentation** -- COCIA'26 Congress (Casablanca)
- **Participation** -- Hackathon RallyIA (2026)

LANGUAGES & MISCELLANEOUS

Arabic : Native **French :** DELF B2 **English :** EFSET B2
Driving License : B **Hobbies :** Reading, Chess, DIY Electronics